Chapter 9, Problem 6P

Problem

For the following pairs of sinusoids, determine which one leads and by how much.

(a) $v(t) = 10\cos(4t - 60°)$ and
$i(t) = 4\sin(4t + 50°)$

(b) $v_1(t) = 4\cos(377t + 10°)$ and
$v_2(t) = -20\cos 377t$

(c) $x(t) = 13\cos 2t + 5\sin 2t$ and
$y(t) = 15\cos(2t - 11.8°)$

Step-by-step solution

Step 1 of 4

(a)

$v(t) = 10\cos(4t - 60°)$

$i(t) = 4\sin(4t + 50°)$

$i(t) = 4\cos(4t + 50° - 90°)$

$i(t) = 4\cos(4t - 40°)$

Rewriting $v(t)$, we have

$v(t) = 10\cos(4t - 40° - 20°)$

This shows that $v(t)$ lags i(t) by $20°$ Or $i(t)$ leads $v(t)$ by $20°$.

Step 2 of 4

(b)

$v_1(t) = 4\cos(377t + 10°)$

$v_2(t) = -20\cos(377t)$

$v_2(t) = 20\cos(377t + 180°)$

Rewriting $v_2(t)$,

$v_2(t) = 20\cos(377t + 10° + 170°)$

Thus shows that $v_2(t)$ leads $v_1(t)$ by $170°$.

Step 3 of 4

(c)

$x(t) = 13\cos 2t + 5\sin 2t$

$y(t) = 15\cos(2t - 11.8°)$

$x(t) = 13\cos 2t + 5\cos(2t - 90°)$

$x(t) = 13\angle 360° + 5\angle -90°$

$x(t) = 13(1 + j0) \ + \ 5(0 - j1) \qquad \rightarrow \quad (1)$

Step 4 of 4

$x(t) = 13 - j5$

$x(t) = 13.928\angle -21.037$

$x(t) = 13.928\cos(2t - 21.037)$

$y(t) = 15\cos(2t - 11.8°)$

Rewriting $x(t)$, we have

$x(t) = 13.928\cos(2t - 11.8° - 9.237°)$

This shows that x(t) lags y(t) by $9.24°$